IAN TAI AHN

Software & ML Engineer

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SUMMARY

Software & ML Engineer with an M.S. in Data Science and 3+ years of professional experience building production systems in high-compliance environments. Trained NLP models (BERT, RoBERTa, sentence-transformers) using PyTorch and scikit-learn; built custom RAG architectures with chunking, embedding, and retrieval pipelines; and deployed end-to-end ML workflows on AWS (EC2, SageMaker, Lambda). Currently serves as Technical Lead and Scrum Master for a mission-critical platform spanning 30+ distributed microservices. Independently builds healthcare ML tooling, including a Medicare prior authorization rules engine grounded in real LCD policy and ICD-10/CPT coding logic. Active Secret clearance.

TECHNICAL SKILLS

Languages & Data: Python, SQL, PostgreSQL, R, Java, JavaScript, TypeScript, Shell, Linux, Snowflake, dbt

Data Engineering: pandas, NumPy, ETL/ELT pipelines, data cleaning, feature engineering, data ingestion

ML Frameworks: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, sentence-transformers, Keras, Spacy

NLP & AI: BERT, RoBERTa, transformer fine-tuning, custom RAG architectures, LLM APIs (Anthropic Claude, Groq, OpenAI), prompt engineering, structured output generation, Pydantic, LangChain

Cloud & Infra: AWS (EC2, S3, SageMaker, Lambda, CloudWatch), Azure, Docker, Kubernetes, Helm, Istio, Ansible, Kyverno, Grafana, Splunk

MLOps & DevOps: CI/CD pipelines, GitHub Actions, Helm charts, Robot Framework, Flux, JIRA, Confluence

Frameworks & Tools: FastAPI, React, Next.js, Django, Flask, Spring Boot, REST APIs, Git, Agile/Scrum

Healthcare Domain: PHI handling, HIPAA-aligned system design, Medicare Advantage workflows, prior authorization automation, ICD-10/CPT coding logic

EXPERIENCE

Lockheed Martin — Hill Air Force Base, UT Aug 2023 – Present

Technical Lead & Scrum Master — LODEM Modernization & File Sync | Feb 2026 – Present

- Lead Agile ceremonies, sprint planning, and backlog prioritization as Scrum Master while continuing to contribute technically and mentor developers across a cross-functional team.
- Own spike planning, stakeholder alignment, and sprint execution for a file synchronization initiative across 30+ microservices on Lockheed infrastructure, using PowerShell, Bash, and open-source tooling.
- Coordinate cross-functionally with Unity and systems modeling teams to align delivery across workstreams; proactively clear blockers and translate ambiguous requirements into sprint-ready work.

Software Engineer / Associate Technical Lead — LODEM Modernization | Aug 2025 – Feb 2026

- Delivered full refactor of the React component library for a virtual field damage and repair entry application on the F-35 Lightning II program, eliminating duplicated components and establishing a maintainable architecture.
- Contributed across the full stack including React frontend, Unity WebGL via react-unity-webgl, and PostgreSQL backend while onboarding developers and driving high-complexity cross-functional coordination.

Platform Engineer — CS3 Kubernetes Service Mesh | Apr 2024 – Aug 2025

- Refactored Helm charts and authored Robot Framework test scripts to surface Kyverno policy deadlocks and validate enforcement behavior across a Kubernetes service mesh; managed pod lifecycle testing and supported CI/CD pipeline operations using Flux.
- Maintained service mesh health across routing, logging, monitoring, and cybersecurity compliance using Prometheus, Splunk, and Grafana; wrote Ansible playbooks to validate logrotate functionality within Kubernetes nodes.

Computer Scientist / Test Automation Engineer — F-35 OBPHM | Aug 2023 – Apr 2024

- Built Python data pipelines and automated test frameworks for F-135 turbofan engine health monitoring, transforming multi-table relational data across regression cycles and reducing a 30-minute manual process to seconds (~10 hours/month saved).

PROJECTS

Prior Authorization Policy Compiler | Python, FastAPI, Groq LLM, Pydantic, custom RAG (chunking, embedding, retrieval)

https://github.com/IanTaiAhn/pauth_rc

- Built a custom information extraction pipeline that ingests unstructured Medicare LCD policy documents, chunks and embeds content using sentence-transformers, and retrieves relevant passages to compile structured, ICD-10/CPT-coded prior authorization checklists with requirement logic and denial-prevention guidance.

- Designed a schema-driven architecture using Pydantic with a two-step extraction pipeline to lock checklist structure before populating clinical specifics, reducing LLM hallucination risk in a regulated context.
- Deployed against Medicare LCD L36007 (Lower Extremity Major Joint Replacement) with production evals validating coverage-criteria accuracy, surfacing hard-stop contraindications used by clinic billing teams to reduce PA denials.

Health Workforce Intelligence Dashboard | React, FastAPI, FRED API, SARIMAX — labor market forecasting tool for Utah healthcare orgs
https://github.com/IanTaiAhn/healthcare_dashboard

- Built a full-stack labor market intelligence tool for Utah healthcare organizations using React, FastAPI, and SQLite, integrating FRED macroeconomic data series via API with automated ingestion, quality validation, and daily refresh scheduling.
- Designed a SARIMAX-based time series forecasting pipeline with walk-forward backtesting, stationarity analysis (ADF testing), and automated feature engineering targeting 6-month employment and quit-rate forecasts for the Utah healthcare sector; model training pipeline in progress.
- Engineered a Composite Stress Score (CSS) derived from JOLTS leading indicators, wage pressure metrics, and employment trends, surfacing a single actionable market signal (Tightening / Stable / Easing) for decision-makers across six interactive dashboard panels.

YouTube Sentiment Analysis — M.S. Thesis | Python, PyTorch, BERT, RoBERTa, sentence-transformers, scikit-learn, AWS (EC2, SageMaker, Lambda)

- Designed and deployed an end-to-end NLP pipeline on AWS (EC2, SageMaker, Lambda) that ingested YouTube comments via API, encoded them using sentence-transformers, and applied BERT/RoBERTa-based classification to predict positive/negative/neutral sentiment.
- Ran regression analysis on transformer-encoded comment embeddings using scikit-learn; trained and evaluated models with PyTorch, surfacing sentiment trends with data visualizations built for interpretability.

EDUCATION

M.S., Data Science *(Graduate Certificate, Computational Data Science & ML)*
Weber State University | GPA 3.85 | 2025

B.S., Computer Science | Weber State University | GPA 3.93 | 2023